

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 09)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1
---	---

Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0	1
---	---

1

Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

0	1
---	---

2

Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

0 1 . 3

State **two** changes that would decrease the cut-off ore grade.**[2 marks]**

1

2

10

0 2 . 1

Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.**[1 mark]**

- A Elemental (liquid) mercury
- B Inorganic mercury compounds
- C Organic (methyl) mercury
- D Mercury vapour

☐☐☐☐

0 2 . 2

Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

0 2 . 3

Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

5

03 . 1

Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

03 . 3

Explain why desalination of seawater is only used in some countries.

[2 marks]

1

2

3

4

5

6

7

8

9

10

11

12

13

03 . 4

Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

1

2

3

4

5

6

7

8

9

10

11

12

13

10

0 4

The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s^{-1}	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0 4

. 1

Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer

times

0 4

. 2

Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 4 . 3

Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

1

2

0 4 . 4

Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

10

0 5

Table 1 lists some techniques used in ecological monitoring.

0 5 . 1

Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

.....

.....

.....

.....

.....

.....

.....

.....

0 5 . 2

Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

.....

.....

.....

.....

0 5 . 3

Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1

.....

2

.....

10

0 6

Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / $\times 10^{15}$ dm^3	Annual transfer (inflow = outflow) / $\times 10^{15} \text{ dm}^3 \text{ yr}^{-1}$	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 6**1**

The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]**Answer**

days

0 6**2**

Calculate the residence time of water in the oceans.

[1 mark]**Answer**

years

0 6**3**

Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

06 . 4

Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

06 . 5

Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

10

07

Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

07 . 1

Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer

%

07 . 2

Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

Ultraviolet (UV)

.....

.....

.....

Visible light

.....

.....

.....

Infrared (IR)

.....

.....

.....

07 . 3

Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

.....

.....

.....

.....

07 . 4

Explain the term **thermal stratification** of the atmosphere.

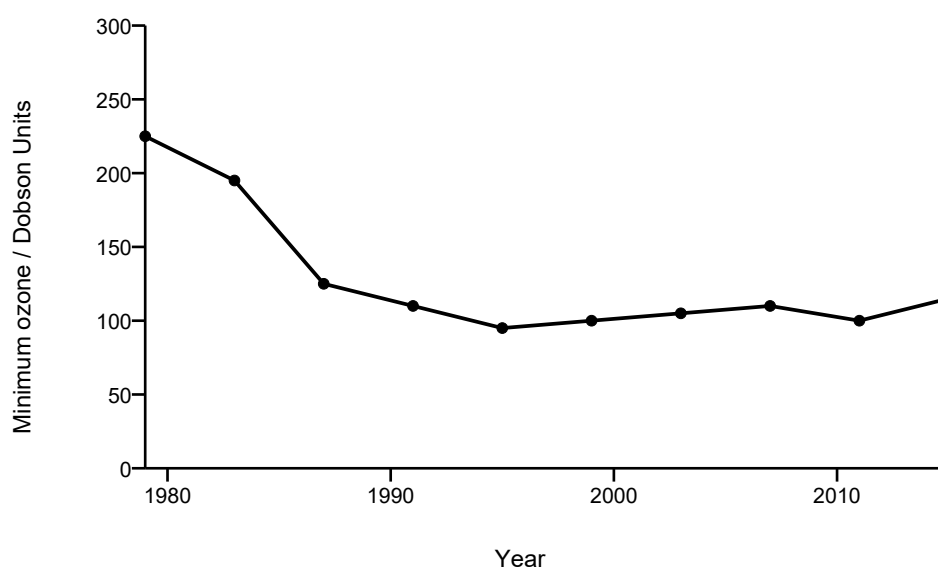
[1 mark]

10

08

Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

Figure 1



08

1 Describe the trend shown in Figure 1.

[2 marks]

08

2 Calculate the percentage decrease in minimum ozone between 1979 and 1995.

[2 marks]

Answer

%

08 . 3

Explain why ozone depletion is most severe over Antarctica in spring.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

08 . 4

Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

.....

.....

.....

.....

10

09

Table 1 shows typical noise levels of some sounds.

Table 1

Sound	Noise level / dB
Threshold of human hearing (at 1000 Hz)	0
Leaves rustling	20
Library	40
Normal conversation	60
Busy road	80
Chain saw	110
Jet engine at 30 m	140

09 . 1

Explain why a logarithmic scale is used to measure sound.

[2 marks]

.....

09 . 2

Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

[2 marks]

Answer

times

09 . 3

A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

09 . 4

Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

09 . 5

State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

10

10 . 1

Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

Extra space

10 . **2** Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2

Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

[illegible]

Do not write outside the box

END OF QUESTIONS